1.18 Graph Sketching: Rational Functions

Worked Solutions

1)

$\frac{5}{6} = \frac{k}{3} \Rightarrow k = \frac{5}{2}$

2 a)

$y = \frac{1}{x-2}$

2 b)

$y = \frac{1}{x} - 1$

2 c)

$y = \frac{1}{x-3} - 2$

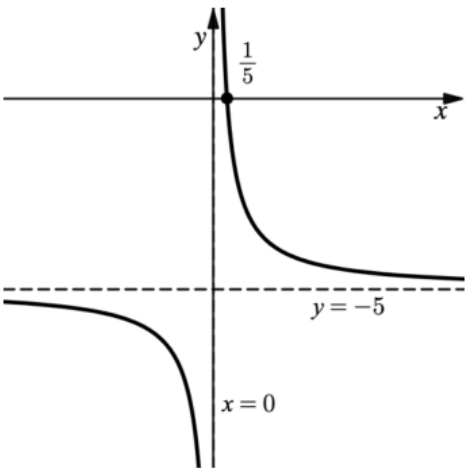
2 d)

$y = \frac{1}{x+2} + 4$

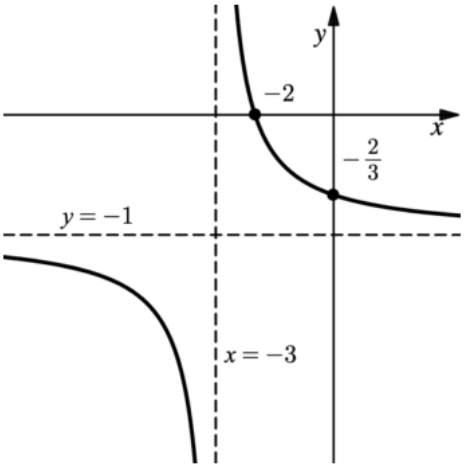
3)

$x = 1$ and $y = 3$

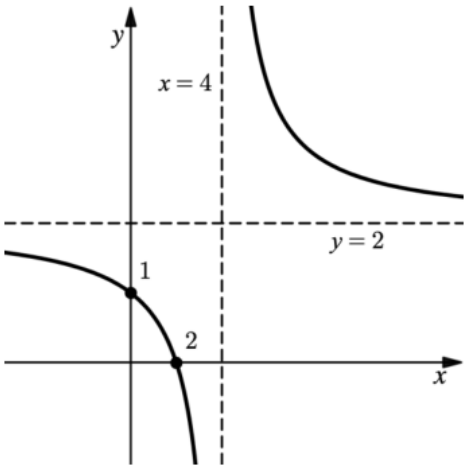
4 a)



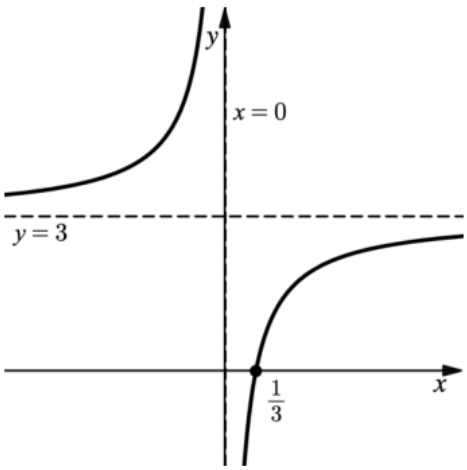
4 b)



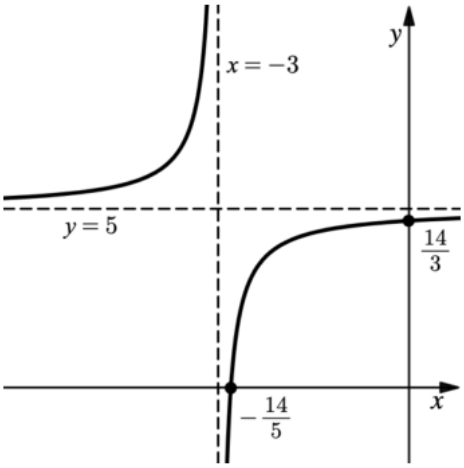
4 c)



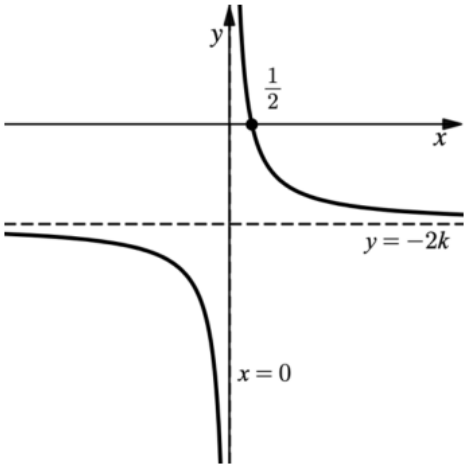
4 d)



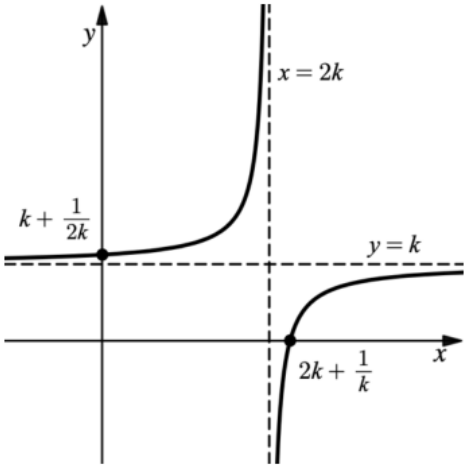
4 e)



5 a)



5 b)



6)

$\frac{6}{5} = \frac{k}{(\frac{1}{2})^2} \Rightarrow k = \frac{3}{10}$

7 a)

$y = \frac{1}{(x+3)^2}$

7 b)

$y = \frac{1}{x^2} - 4$

7 c)

$y = \frac{1}{(x+2)^2} - 1$

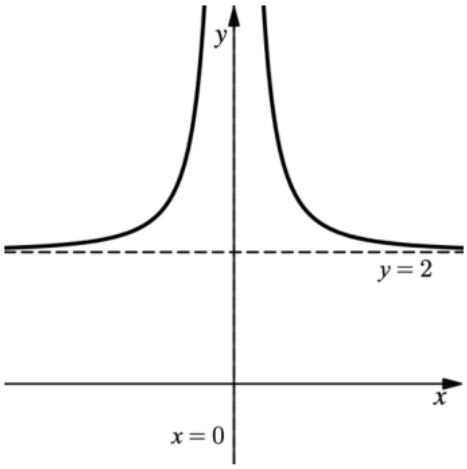
7 d)

$y = \frac{1}{(x-3)^2} + \frac{3}{4}$

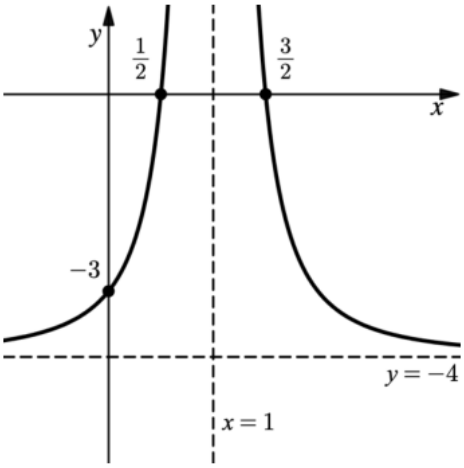
8)

$x = -5$ and $y = -2$

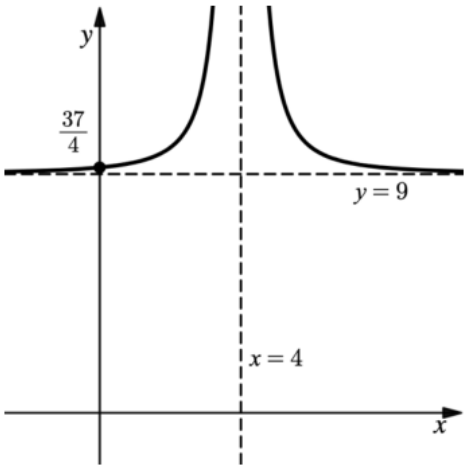
9 a)



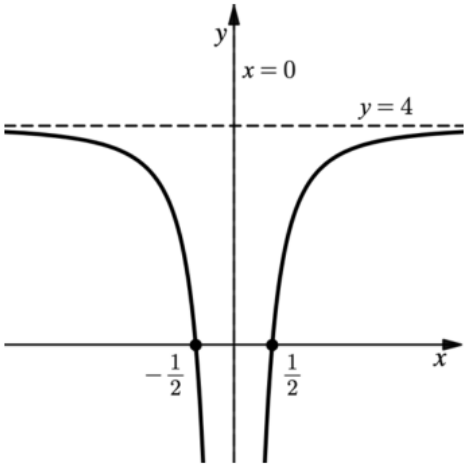
9 b)


y
$\frac{1}{2}$
$\frac{3}{2}$
x
−3
$y = -4$
$x = 1$

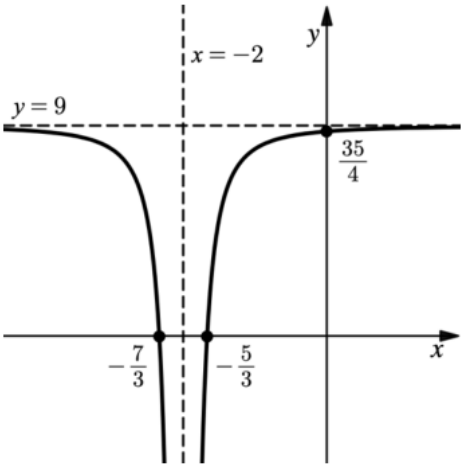
9 c)


y
$\frac{37}{4}$
$y = 9$
$x = 4$
x

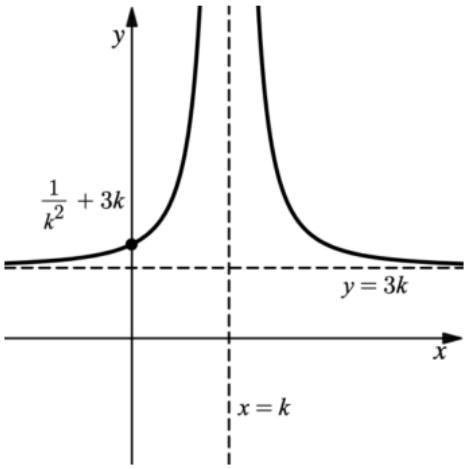
9 d)


y
$x = 0$
$y = 4$
x
$-\frac{1}{2}$
$\frac{1}{2}$

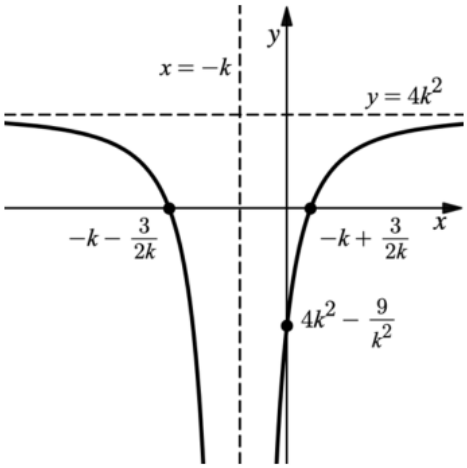
9 e)


$x = -2$
y
$y = 9$
$\frac{35}{4}$
$-\frac{7}{3}$
$-\frac{5}{3}$
x

10 a)


y
$\frac{1}{k^2} + 3k$
$y = 3k$
x
$x = k$

10 b)


y
$x = -k$
$y = 4k^2$
$-k - \frac{3}{2k}$
$-k + \frac{3}{2k}$
x
$4k^2 - \frac{9}{k^2}$